

HITESH GUPTA

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About Me

Passionate Artificial Intelligence and Machine Learning student with hands-on experience in building intelligent systems across computer vision, deep learning, and predictive modelling. Strong technical foundation in Python, NumPy, pandas, Scikit-learn, and OpenCV, combined with practical experience in object detection, image processing, and generative adversarial networks. Committed to continuously evolving as an AI practitioner and applying cutting-edge machine learning techniques to solve real-world challenges.

Education

Vellore Institute Of Technology

B.Tech in Computer Science and Engineering (AI & ML)

Sep. 2023 – May 2027

CGPA: 8.64

DS Science Academy

Class XII

July 2021 – April 2022

Percentage: 89.80%

Experience

Edunet Foundation (In association with Microsoft and AICTE)

June 2025 – July 2025

AI and Machine Learning Intern

Remote

- Gained hands-on experience with core ML concepts including supervised and unsupervised learning, classification, regression, and unlabeled data handling through practical labs on Microsoft Learn and Edunet LMS.
- Explored Generative AI and Computer Vision on Microsoft Azure, working with Azure Custom Vision for image classification and real-world Gen AI applications through practical demos.
- Strengthened understanding of Neural Networks and Deep Learning, covering training pipelines, convolutional layers, and optimization techniques through structured hands-on sessions.
- Developed “**AI-Based Medical Image Enhancement for Diagnostics Using GANs**” – a GAN-based system to enhance low-resolution medical images, improving diagnostic accuracy in resource-constrained healthcare settings.

Technical Skills

Languages: Python, Java, SQL

Developer Tools: Git, GitHub, VS Code, Jupyter Notebook, Google Colab

Technologies/Frameworks: NumPy, pandas, Scikit-learn, Matplotlib, OpenCV, TensorFlow

Projects

Human Sustainability Score Prediction Model | *Python, Scikit-learn, pandas, NumPy, Matplotlib*

- Designed an AI-driven regression pipeline to predict individual sustainability scores from lifestyle data including energy use, transport, and recycling habits.
- Applied feature engineering and modular ML modeling for scalable and interpretable predictions across diverse user profiles.
- Visualized key data insights using Matplotlib to support eco-friendly behavior analysis and data-driven policymaking.

Lane Detection and Driver Assistance System | *Python, NumPy, OpenCV, Image Pre-processing, ROI Masking*

- Built a real-time lane detection pipeline using OpenCV with image preprocessing, Gaussian blur, and Canny edge detection for accurate lane identification.
- Applied ROI masking and Hough Transform to detect and overlay lane markings on live video frames.
- Designed for ADAS and autonomous driving use cases, ensuring reliable detection under varying road and lighting conditions.

Smart Retail Checkout System | *Python, YOLOv8 (Ultralytics), OpenCV, Matplotlib, Google Colab*

- Developed an AI-powered retail checkout system detecting grocery products from images using a pre-trained YOLOv8 Nano model via the Ultralytics library.
- Implemented a modular pipeline for object detection, class label extraction, and price dictionary lookup to automatically generate itemized bills.
- Designed the architecture to be extensible toward automated bill generation and real-time inventory management.

Achievements

Certifications:

- Applied Machine Learning in Python (Coursera)
- Java Programming (HackerRank)
- Data Analytics Job Simulation (Deloitte Virtual Experience Program)

Competitive Programming:

- Solved **500+** algorithmic problems across LeetCode, CodeForces, and GeeksforGeeks.